

CSI132 - DISCRETE STRUCTURES II

Tutorial on Relations and Graphs

ATTEMPT ALL QUESTIONS

DURATION: 2 HOURS

1. Using the information below about course final exams at a University, answer questions that follow.

The courses to be scheduled for final exam are: MAT115, MAT116, MAT185, MAT195, CSI101, CSI102, CSI273 and CSI473. There are **NO** students taking both:

- MAT115 and CSI473
- MAT116 and CSI473
- MAT195 and CSI101
- MAT195 and CSI102
- MAT115 and MAT116
- MAT115 and MAT185
- MAT185 and MAT195

They are students in **EVERY** other pair of courses.

- (a) Draw a conflict graph to represent the above problem.
- (b) Give a proper coloring of the graph using the smallest number of colors
- (c) Give an adjacency list representation of the graph.
2. Six stations are located at distances shown in the table below. Two stations cannot use the same same channel when they are within 150 miles of each other.

Table 1: assessment weights

	1	2	3	4	5	6
1	–	85	175	200	50	100
2	85	–	125	175	100	160
3	175	125	–	100	200	250
4	200	175	100	–	210	220
5	50	100	200	210	–	100
6	100	160	250	220	100	–

- (a) Draw a conflict graph to represent the above problem.

- (b) Give a proper coloring of the graph using the smallest number of colors
- (c) Give an adjacency list representation of the graph.
3. Consider a relation represented by the graph below:

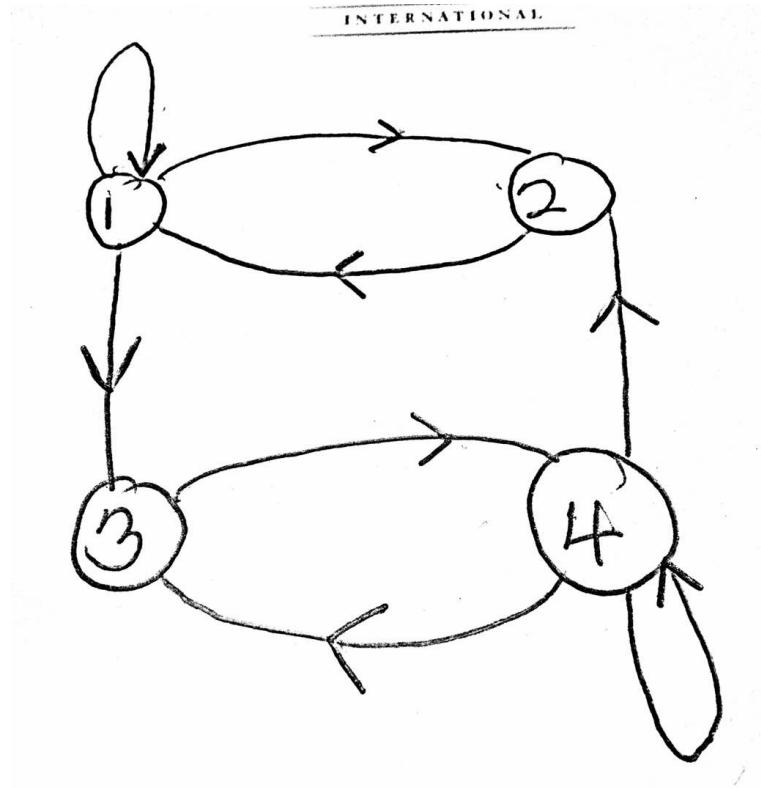


Figure 1: Graph representing a relation.

- (a) Is the relation reflexive?
- (b) Is the relation symmetric?
- (c) Is the relation transitive?
- (d) List the ordered pairs in this relation.
4. Consider the relation $R = \{(a, b) | a \text{ divides } b\}$ over set $A = \{1, 2, 3, 4, 6\}$
- (a) Is this relation reflexive?
- (b) Is this relation symmetric?
- (c) Is this relation transitive?
- (d) Represent the relation using a graph.